Dimming, orbits, clocks and abundances: archival tests for technosignatures among the stars

Charles C. Norton

Abstract

A technology at a star could change the star's light, the motions of the bodies around it, the timing of its variations or the composition of its surface, in ways that its astrophysics does not. We test each kind of change against the archive that measures it. In light, no star dims over a century, between Hipparcos and Gaia or within Gaia's epochs without a natural cause, no transit deepens from Kepler to TESS in the light curves themselves, the Kepler giants lack flux as often as they gain it, no variable changes colour while holding its brightness or holds its brightness at discrete levels, and no hot body is smaller than a white dwarf. In motion, no orbit is too light for the light it shows, no eclipsing binary too light for its eclipses, no transit faster than escape velocity, no planet pair in exact commensurability, no orbit synchronised with its star beyond the tidal range, no planet aimed at the Earth, and no planetary system or co-moving group arranged beyond chance. In timing, no two stars share a clock beyond the period distribution, and no star in the Earth's transit zone or on the ellipsoid of a nova flashes at the moment it would see us or the nova. In composition, no dwarf carries fission products or processed carbon and nitrogen without a natural cause. The one result that remains open, a 5.5 per cent polarisation assigned to an unreddened star in 1977, is the subject of its own note.

Keywords: Technosignatures, Stars: variables: general, Binaries: general, Stars: abundances, Astrometry

1. Introduction

Most technosignature searches look for a signal; a technology also leaves the star it occupies changed. Material in orbit dims and reddens the starlight it intercepts and re-radiates it in the infrared [1, 2], a structure placed to be seen transits in a way a planet does not [3], and material deposited in a star changes its surface composition [4]. The large surveys of the last three decades measure every star near the Sun in brightness, colour, position, motion and composition, at many epochs, and each of these changes has a limit the star's own physics sets. This paper puts the limits to the surveys, with a null drawn from the same data for each, and states what remains.

1.1. Contributions

1. Tests of starlight: dimming over a century, between Hipparcos and Gaia and within Gaia's epochs, transit depths from Kepler to TESS, missing flux in asteroseismic giants, colour without brightness and brightness at discrete levels, hot bodies smaller than white dwarfs, and dust at white dwarfs.

2. Tests of motion: orbits too light for their light, eclipsers too light for their eclipses, transits faster than escape velocity, exact commensurabilities, synchronous orbits, aimed transits, the orientation of orbits and the clustering of co-moving stars.

3. Tests of timing and geometry: shared clocks, the Earth's transit zone at opposition, the ellipsoids of novae and the stars in the beams of planetary radar.

4. Tests of composition: fission products and processed carbon and nitrogen in dwarfs.

2. Light

Dimming over a century. A structure built over a century would dim its star by a measurable amount. In the Harvard plates of DASCH [5], the 300 nearest sun-like stars of $G = 8$ to 11 have differential trends against comparison stars on the same plates with a spread of 0.12 mag per century, and the two above 3.5σ have slopes in their plate series that disagree with one common slope; injected fadings are recovered in 7, 60 and 83 per cent of cases at 0.4, 0.6 and 1.0 mag per century.

Dimming between Hipparcos and Gaia. For 15,189 Hipparcos stars within 100 pc fainter than $G = 6$, Gaia's G against the Hipparcos magnitude [6] after a colour term spreads by 0.0034 mag for constant single stars; 218 stars are fainter in Gaia by five robust deviations, every one with a natural cause: a Hipparcos variability flag, a neighbour Hipparcos summed, a close companion that Gaia's G misses, a spotted rotator or an astrometric binary. Of the 531 brighter in Gaia, 19 have none.

Dips within Gaia and Hipparcos epochs. Of 130 variables within 100 pc whose Gaia epoch photometry dips with a skewness above 1.5 outside the classifier's natural classes, every dipper is a single epoch, a single transit with its neighbours undimmed, a close binary measured differently at each scan angle, a periodic eclipse, reddening or a dip in G alone. Of 17,285 Hipparcos stars, 23 keep a dip seen in both fields of view, the rate beyond 100 pc, and the deepest is an eclipsing binary in TESS. The 28 Gaia alerts within $2''$ of a star within 100 pc are flares on red and brown dwarfs, two dwarf-nova outbursts, a young star's flare, a flare star's slow rise and a distant dwarf nova behind a spurious parallax. Of the stars with both a Hipparcos-to-Gaia deviation and a NEOWISE W2 trend, 671, the highest combined value of optical dimming and mid-infrared brightening arises by chance at its rank 99 per cent of the time against the W2 trends re-paired among stars of like brightness.

Transit depths from Kepler to TESS. A structure under construction, or a swarm filling in, deepens its transit at the same period and duration. Of 102 Kepler objects of interest that are TESS objects of interest within $6''$ and 0.2 per cent in period, three are deeper in TESS at 5σ with the duration held, and two shallower, the null. Measured the same way in both missions' light curves, each transit's centre against a line fitted to its sides, two of the deeper transits are deeper only in the crowding-corrected flux, the pipeline's estimate of the neighbours' light in TESS's $21''$ pixels, and one is deeper only in the TOI table; of the shallower, one is diluted by light in the aperture beyond the crowding model and one is shallower only in the table.

Missing bolometric flux. A star's seismic radius and temperature give its luminosity; a structure intercepting its light leaves Gaia's luminosity short of it. Among 14,959 single Kepler giants with seismic radii [7] and Gaia luminosities [8], deficits are as common as excesses at every threshold, 31 against 35 beyond 4σ, and the extreme deficits have a dwarf's luminosity and a giant's oscillations: background giants in foreground dwarfs' apertures.

Colour without brightness, and brightness in levels. A filter that changes a star's colour while holding its brightness would be no spot or companion. Of 1073 Gaia variables brighter than $G = 16$ whose G scatter is below 0.003 mag while BP and RP scatter by 0.02 or more, 899 exceed the quiet sources' tail; the 60 highest, read epoch by epoch, move BP and RP together, a neighbour entering both windows, or scatter in one band alone, and none moves BP against RP. Conversely, of 1845 ZTF spotted variables [9], nine have an amplitude ratio that is grey at 3σ: eight RS CVn and BY Dra systems above the main sequence, where the companion dilutes the spots' colour, and a pre-main-sequence star in NGC 2264 occulted by its disk. In the TESS 20 s light curves of the stars within 30 pc, 2096 steps at 7σ with transitions of two minutes or less are, 149 of them, common to three or more stars of a sector, and the rest split 1484 up and 460 down, flare rises; no light curve holds three levels each revisited.

Bodies too small or too blue. A body hot by colour and smaller than a white dwarf would need a source other than a star. Among 5278 sources within 100 pc with black-body colour temperatures above 7000 K, 22 with clean astrometry and photometry lie below $0.003\,R_\odot$, all of them catalogued white dwarfs, four in the list of infrared-faint white dwarfs of Bergeron et al. [10], whose collision-induced absorption makes a 3000 K photosphere read hot by colour, and none with an ultraviolet detection. In the Euclid Q1 catalogue, none of the 14 point sources bluer than the Rayleigh-Jeans slope in $I_E - H_E$ is so at 3σ with clean flags, unblended and steady.

White dwarfs. Dust at a white dwarf pollutes its atmosphere with the metals it accretes, and warm material without pollution would be no disc of rock. Of 1434 DESI DR1 white dwarfs with unWISE W1 and W2 and parallaxes at 5σ, 13 exceed the photosphere in both bands, five of them unblended with excesses of 900 to 1500 K and 1.6 to 3.3 mag in W1, the excess of an unresolved brown dwarf rather than the 0.3 to 1 mag of a dust disc. Of 16,676 white dwarfs within 100 pc [11], 25 have W3 or W4 above the photosphere with W1 and W2 photospheric; the strongest are an extended source in the Galactic plane, an optical ghost, single-exposure detections and cirrus.

3. Motion

Light without mass. A luminous object that weighs nothing moves its star's photocentre and not its spectrum. Of 42 Gaia DR3 astrometric orbits within 100 pc with HARPS or HIRES velocities, nine have photocentre orbits the velocities exclude: close and long-period companions, a period beside Gaia's scanning law, and blended double-lined stars, two of which Gaia itself resolves on part of its transits.

Eclipsers too light for their eclipses. An opaque screen eclipses without moving its star. Among 956 Gaia eclipsing binaries deeper than 0.75 mag with ten or more RVS transits, the observed velocity amplitude over the least a main-sequence eclipser of the depth's size demands has a median of 0.73, and the ten below 0.1 are a saturated giant, an unphysical model depth, two binaries whose blended lines read low, a Kepler system whose true period and depth differ from Gaia's, and five stars flat at Gaia's period in ZTF and TESS: Gaia's depths and periods, not the velocities, fail.

Transits faster than escape velocity. A bound body crosses its star no faster than the star's surface escape speed, which sets a floor on a full transit's duration. Of the Kepler threshold-crossing events with characterised hosts [8], 281 fall below 0.8 of the floor at their fitted geometry: 253 on giants, 26 pulsators, an eclipsing binary, a KOI with centroid flags and a single cadence flagged for cosmic rays. Of the TESS objects of interest, 242 fall below the floor at zero impact parameter: giants, contact binaries and pulsators, published false positives, and 42 candidates without an impact parameter in the catalogue. Their TESS light curves, folded at the catalogue

ephemeris and fitted with a trapezoid whose flat-bottom fraction bounds the impact parameter, give 20 a duration that a bound orbit reaches at some impact parameter, four a V shape that leaves a grazing chord, and three a flat bottom whose largest allowed impact parameter keeps the crossing within the escape speed; five show no transit at the catalogue ephemeris, and ten have no TESS light curve in the public archives.

Commensurability and synchrony. An engineered system could hold its planets in exact resonance or its orbits in step with its star. Of 2393 planet pairs, none lies within 10^{-5} of a small-integer ratio, and five within 10^{-4} against 3.4, in the librating chains of TOI-1136, Kepler-223 and HD 110067. Of 541 Kepler planets at periods of 10 d or more around hosts with rotation periods [12, 13], 25 orbit within 2 per cent of the rotation period, its double or its half, against 23.4 ± 4.6 with the periods shuffled, and the four 1:1 matches drift in rotational phase by 0.17 to 0.25 of a cycle over four years, as spots do.

Aimed transits. A structure placed to be seen from the Earth transits at a small impact parameter. The fitted impact parameters of 2294 Kepler candidates and planets pile up at zero where the fits of injected planets do too, through poorly constrained fits at the parameter's lower bound, and the 153 planets with impact parameters measured to 0.1 hold 7 below 0.05 against 11.3 expected.

Orbits and groups. Of 1361 Gaia astrometric companions that may be substellar [14], 6.2 per cent have an orbital pole within 15° of a Galactic pole against 6.3 per cent of the stellar companions at the same latitudes; the 74 whose upper mass bound lies below $0.2\,M_\odot$ hold nine against 4.7 ± 2.1 ($p = 0.04$), one reading of a small set among several. Of 6044 APOGEE dwarfs within 100 pc with velocities, the 31 groups within 10 pc and $1\,\mathrm{km\,s^{-1}}$ are the Hyades and co-natal pairs, with metallicity spreads only at the pipeline's temperature edge; and the 2977 wide binaries of the coverage store share no anomaly between their members beyond the permutations.

Imaging candidates. Of the candidate companions of the Gemini/NICI and NaCo direct imaging surveys with two or more epochs, the few off both the co-moving and the background track are known brown dwarfs in orbital motion, a background star on its track in Gaia DR3, and a faint source moving $200\,\mathrm{mas\,yr^{-1}}$ of its own, as a field brown dwarf would.

4. Timing and geometry

Shared clocks. Beacons built by one civilization could share a period. Among 770,043 ZTF periodic variables [9], pairs more than 10″ apart that agree in period scale linearly with the tolerance, as a smooth period distribution gives, and those that agree in phase as well number 42,112 against 41,384 to 42,020 with the phases shuffled; the largest groups sit on periods the catalogue repeats exactly.

The Earth's transit zone at opposition. A star in the Earth's transit zone sees the Earth transit the Sun at our opposition, the moment a signal timed to that event would leave. Of the stars within 100 pc and 0.27° of the ecliptic with TESS two-minute data at opposition, 27 windows of sectors 42 to 46 hold no lone flash against 0.37 expected and no brightening against 1.12, and 12 windows of sectors 70 to 72 hold one lone flash against 0.53.

Nova ellipsoids. A civilization that saw a nova could signal when its light reached them, and stars on the ellipsoid with the nova and the Earth at its foci would be seen to do so now [15, 16]. Of the stars within 100 pc on the ellipsoids of T CrB's 1866 and 1946 eruptions, GK Per, V603 Aql, RR Pic, DQ Her, CP Pup, V1500 Cyg and the supernovae of 1572 and 1604, observed by TESS

at their arrival, the flashes and light curves with flashes match the same stars' other sectors, the one excess carried by active binaries whose flares follow the sectors.

Radar beams and beams between stars. A star in the beam of a planetary radar would reply after twice its light-travel time. The 813 stars within 100 pc in the beams of every published radar measurement from 1968 to 2025 are no closer than 48.9 light years, so no reply is due before 2117. And of the 2705 stars within 20 pc, 37 pairs lie on opposite sides of the Earth within a 0.25° beam, as many as random directions give.

5. Composition

Fission products. Fission waste deposited in a star raises zirconium, molybdenum, ruthenium, barium, neodymium and europium together in a pattern no nucleosynthesis makes [4]. Among 50,742 GALAH DR4 stars [17] with clean abundances of the six, fission-like patterns are rarer than the mirror pattern at every threshold, 71 against 230 above 5σ, and the 23 FGK dwarfs with molybdenum and ruthenium all lie within 2.5σ of solar.

Processed carbon and nitrogen. A dwarf whose surface carries the products of the CNO cycle has gained them from a companion. Of 31,351 APOGEE dwarfs, 91 are nitrogen-rich at 5σ against the locus in temperature and metallicity and none carbon-rich; 57 of the 91 are Gaia binaries against 28 per cent of random dwarfs, the mark of transfer from an evolved companion, 78 carry binary, s-process, sodium, rotation, evolution or temperature-edge flags, and the 13 that carry none all carry the pipeline's microturbulence grid-edge warning, which biases carbon and nitrogen in dwarfs.

6. Related work

Dyson [1] proposed the infrared signature of a star's surroundings reprocessed, and Wright et al. [2] set out the search for it; Arnold [3] proposed artificial transits, Whitmire and Wright [4] fission waste in a star, and Davenport et al. [15, 16] searched the ellipsoid of SN 1987A for signals timed to it. The tests here rest on the surveys' own catalogues: Hipparcos and Gaia [6, 14], DASCH [5], the Kepler giants' asteroseismology [7] and luminosities [8], ZTF's periodic variables [9], the Kepler rotation periods [12, 13], the Gaia white dwarfs [11, 10] and GALAH [17].

7. Conclusion

No test finds a star changed in light, motion, timing or composition beyond what its astrophysics and the surveys' own systematics give. Dimming is variability, blending or an epoch's error; deepened transits are crowding corrections and catalogue depths; missing flux is a background giant; colour without brightness is a neighbour; the hot small bodies are infrared-faint white dwarfs; light orbits are blends and companions; light eclipsers are Gaia's depths; fast transits are giants, pulsators and chords away from the centre; commensurabilities are librating chains; aimed transits are the fits' degeneracy; shared clocks are the period distribution; the transit zone and the nova ellipsoids hold flashes at the background rate; and the abundance outliers are binaries and pipeline flags. The reach of each test is its survey's: the century test reaches fadings of about 0.6 mag per century, the Hipparcos-to-Gaia test 0.02 mag over 25 years, the missing-flux test deficits of 45 per cent, and the motion tests the precision of Gaia's astrometry and the archives' velocities.

Table 1: The tests, their samples and results.

Test	Sample	Result
Century dimming	300 DASCH stars	no consistent trend
Hipparcos to Gaia	15,189 stars	every dimming natural
Gaia and Hipparcos dips	130; 17,285 stars	natural, at the background rate
Transit depths	102 KOI-TOI pairs	crowding corrections, catalogue depths
Missing flux	14,959 giants	deficits as common as excesses
Colour without brightness	1073 Gaia variables	neighbours in the windows
Discrete levels	1806 TESS sectors	flare rises and shared steps
Too small or too blue	5278; 14 sources	infrared-faint white dwarfs; flags
White-dwarf dust	1434; 16,676 white dwarfs	brown dwarfs, artefacts
Light without mass	42 orbits	companions and blends
Light eclipsers	956 binaries	Gaia's depths and periods
Escape velocity	Kepler, TESS events	giants, pulsators, false positives, chords
Commensurability	2393 pairs	librating chains
Synchrony	541 planets	at the chance rate
Aimed transits	2294 planets	the fits' pile-up
Orbit poles, groups	1361 orbits; 6044 dwarfs	at chance; co-natal
Shared clocks	770,043 variables	at the period distribution
Transit zone at opposition	39 windows	at the background rate
Nova ellipsoids	ten eruptions	at the background rate
Fission products	50,742 stars	rarer than the mirror
Processed CNO	31,351 dwarfs	binaries and pipeline flags

Acknowledgements

This work uses Gaia DR3 and Hipparcos (ESA; Gaia DPAC), DASCH (Harvard College Observatory), Kepler and TESS through MAST, ZTF through IRSA, NEOWISE and AllWISE (IRSA), APOGEE (SDSS-IV), GALAH DR4, DESI DR1 through the Astro Data Lab, Euclid Q1 (ESA), the NASA Exoplanet Archive and ExoFOP, the Gemini/NICI and VLT/NaCo survey tables, JPL's radar astrometry, and SIMBAD and VizieR at CDS, Strasbourg.

References

[1] F. J. Dyson, Search for artificial stellar sources of infrared radiation, *Science* **131**, 1667 (1960).

[2] J. T. Wright, B. Mullan, S. Sigurdsson and M. S. Povich, The $\hat{G}$ infrared search for extraterrestrial civilizations with large energy supplies. I. Background and justification, *Astrophys. J.* **792**, 26 (2014).

[3] L. F. A. Arnold, Transit light-curve signatures of artificial objects, *Astrophys. J.* **627**, 534 (2005).

[4] D. P. Whitmire and D. P. Wright, Nuclear waste spectrum as evidence of technological extraterrestrial civilizations, *Icarus* **42**, 149 (1980).

[5] S. Tang, J. Grindlay, E. Los and M. Servillat, Improved photometry for the DASCH pipeline, *Publ. Astron. Soc. Pac.* **125**, 857 (2013).

[6] M. A. C. Perryman, L. Lindegren, J. Kovalevsky, et al., The HIPPARCOS catalogue, *Astron. Astrophys.* **323**, L49 (1997).

[7] J. Yu, D. Huber, T. R. Bedding, et al., Asteroseismology of 16,000 Kepler red giants: global oscillation parameters, masses, and radii, *Astrophys. J. Suppl.* **236**, 42 (2018).

[8] T. A. Berger, D. Huber, J. L. van Saders, et al., The Gaia-Kepler stellar properties catalog. I. Homogeneous fundamental properties for 186,301 Kepler stars, *Astron. J.* **159**, 280 (2020).

[9] X. Chen, S. Wang, L. Deng, R. de Grijs, M. Yang and H. Tian, The Zwicky Transient Facility catalog of periodic variable stars, *Astrophys. J. Suppl.* **249**, 18 (2020).

[10] P. Bergeron, P. Dufour, G. Fontaine, et al., On the nature of ultracool white dwarfs: not so cool after all, *Astrophys. J.* **934**, 36 (2022).

[11] N. P. Gentile Fusillo, P.-E. Tremblay, E. Cukanovaite, et al., A catalogue of white dwarfs in Gaia EDR3, *Mon. Not. R. Astron. Soc.* **508**, 3877 (2021).

[12] A. McQuillan, T. Mazeh and S. Aigrain, Stellar rotation periods of the Kepler objects of interest: a dearth of close-in planets around fast rotators, *Astrophys. J. Lett.* **775**, L11 (2013).

[13] T. Mazeh, H. B. Perets, A. McQuillan and E. S. Goldstein, Photometric amplitude distribution of stellar rotation of KOIs: indication for spin-orbit alignment of cool stars and high obliquity for hot stars, *Astrophys. J.* **801**, 3 (2015).

[14] Gaia Collaboration, F. Arenou, C. Babusiaux, et al., Gaia Data Release 3: stellar multiplicity, a teaser for the hidden treasure, *Astron. Astrophys.* **674**, A34 (2023).

[15] J. R. A. Davenport, B. Cabrales, S. Sheikh, et al., Searching the SETI ellipsoid with Gaia, *Astron. J.* **164**, 117 (2022).

[16] B. Cabrales, J. R. A. Davenport, S. Z. Sheikh, et al., Searching the SN 1987A SETI ellipsoid with TESS, arXiv:2402.11037 (2024).

[17] S. Buder, J. Kos, E. X. Wang, et al., The GALAH survey: data release 4, arXiv:2409.19858 (2024).